B.Sc in Computer Science and Engineering Thesis Report on

Poly-Dialectal Neural Machine Translation System for Bangla Regional Dialects

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1 Introduction

1.1 Background and Motivation

Bangla (Bengali), an Indo-Aryan language, stands as one of the most widely spoken languages globally, boasting over 240 million native speakers primarily distributed across Bangladesh and the Indian states of West Bengal, Tripura, and Assam. Despite its immense socio-linguistic prominence and rich literary heritage, Bangla is not a monolithic entity; rather, it exhibits substantial intra-language diversity through a vast array of distinct regional dialects. Dialects such as Sylheti, Chittagonian, Barisali, Noakhali, Rangpuri, Rajshahi, and Mymensingh differ markedly from Standard Colloquial Bangla (SCB) across multiple linguistic strata. These divergences manifest as profound phonological variations, complex morphological inflections, altered syntactic structures, and highly localized lexical choices. In extreme cases, such as the Chittagonian or Sylheti dialects, the linguistic deviation is so severe that they are often bordering on mutual unintelligibility with Standard Bangla.

This deep intra-language divergence engenders significant communication barriers within the broader socio-economic framework. Native speakers of peripheral regional dialects frequently encounter difficulties in seamless communication with speakers of other dialects, as well as in formal interactions, legal proceedings, and educational instructions traditionally conducted in Standard Bangla. Furthermore, modern Natural Language Processing (NLP) tools, machine translation systems, and digital interfaces—ranging from virtual assistants to automated governance platforms—are overwhelmingly optimized exclusively for Standard Bangla. Consequently, millions of dialectal speakers face systemic digital marginalization and language-access inequality, lacking automated communication technologies tuned to their native spoken forms. This technological disparity highlights an urgent need for inclusive NLP systems that can bridge the communication gap without enforcing linguistic homogenization.

1.2 Problem Statement

In recent years, Neural Machine Translation (NMT) architectures and Large Language Models (LLMs) have achieved state-of-the-art results for cross-lingual tasks involving high-resource languages. However, their efficacy degrades severely when applied to low-resource intra-language dialectal shifts. Most existing Bangla NLP benchmarks and pre-trained models treat the language as a singular, uniform distribution, fundamentally ignoring the complex cross-dialectal matrix.

This oversight stems from a confluence of critical technical challenges. Firstly, there is an acute scarcity of multi-dialectal parallel corpora required to effectively fine-tune deep neural networks. Secondly, the lack of standardized orthography for regional dialects exacerbates tokenization mismatches; standard Byte-Pair Encoding (BPE) [17] or Word-Piece [24] algorithms optimized for SCB frequently fragment dialectal words into meaningless subwords, destroying semantic integrity. Finally, existing regional text datasets suffer from severe class imbalance and are predominantly tailored for simple classification tasks rather than complex generative alignment. Addressing these challenges requires a paradigm shift from uni-dialectal modeling to a holistic, poly-dialectal approach.

1.3 Proposed System and Contributions

To address these fundamental limitations, this research introduces a unified **Poly-Dialectal Neural Machine Translation System** tailored specifically for the morphological and syntactic nuances of Bangla regional dialects. As illustrated in Figure 1, our architecture enables robust, multi-directional translation across three core operational paradigms: Dialect → Standard Bangla, Standard Bangla → Dialect, and direct Dialect → Dialect translation, thereby eliminating the cascading errors typically introduced by pivot-based translation methods.

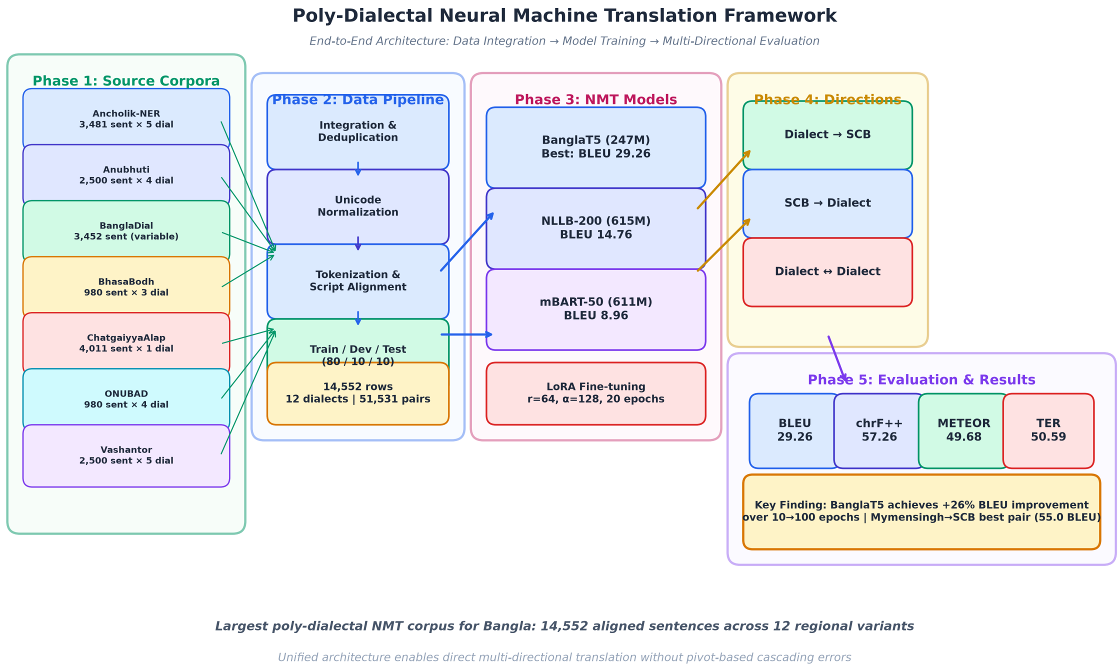

Figure 1: End-to-end architectural workflow of the Poly-Dialectal Neural Machine Translation Framework, depicting five phases: source corpora integration from seven existing datasets, the data processing pipeline, NMT model fine-tuning with LoRA, multi-directional translation paradigm, and evaluation with four standard MT metrics.

The primary contributions of this thesis are summarized as follows:

1. **Largest Multi-Dialectal Parallel Corpus:** We construct the most expansive multi-dialect machine translation corpus for Bangla to date, comprising 14,552 aligned rows across 12 regional dialect columns. The corpus integrates seven prior datasets—Ancholik-NER, Anubhuti, BanglaDial, BhasaBodh, ChatgaiyyaAlap, ONUBAD, and Vashantor. Additionally, we incorporate 500 manually created, bidirectional, and verified high-quality parallel sentence pairs for five dialects (Rangpur, Tangail, Kishoreganj, Narail, Narsingdi), yielding over 51,531 non-null parallel pairs across 11 regional variants.

2. **Pioneering Benchmark of NMT Architectures on Poly-Dialectal Translation:** To the best We exeof our knowledge, this work presents the first systematic empirical evaluation of prominent sequence-to-sequence architectures—specifically BanglaT5 [2], NLLB-200 [22], and mBART-50 [21]—on the novel task of multi-directional, poly-dialectal Bangla translation.

3. **Systematic Cross-Dialectal Linguistic Analysis & Scaling Study:** We conduct a rigorous Optimal Dataset Size Study (progressively scaling from 5,000 instances) and quantitative characterization of cross-dialectal transfer mechanisms. We uncover how morphological proximity and training data volume jointly influence neural machine translation efficacy across diverse metrics.

4. **Real-World Web Deployment:** We deploy the best-performing NMT model in a fully functional web application, bridging the digital language divide by offering real-time poly-dialectal translation capabilities accessible to end users.

1.4 Thesis Organization

The remainder of this report is organized as follows: Section 2 critically reviews related literature on Bangla regional NLP and dialectal machine translation. Section 3 formally states the research hypothesis and objectives. Section 4 details the methodology, including dataset construction, alignment strategies, and model configurations. Section 5 presents a comprehensive evaluation of the experimental results. Section 6 discusses the limitations of the current framework and proposes future research directions, followed by concluding remarks in Section 7.

2 Literature Review

2.1 Overview of Regional Bangla NLP

Historically, the development of linguistic resources for Bangla has heavily favored Standard Colloquial Bangla, leaving regional dialects severely underrepresented in the NLP ecosystem. However, over the past few years, interest in low-resource regional Bangla NLP has seen a significant upsurge. Initial studies focused predominantly on resource creation for text classification and sentiment analysis tasks for specific regions.

Sultana et al. [20] established the *BdRegionText* corpus, a pioneering effort for regional text classification, which starkly demonstrated the performance bottleneck of standard classifiers when exposed to rich dialectal vocabulary. Building on the need for broader representation, Mahi et al. [11] released *BanglaDial*, a merged text dataset that brought to light the severe class imbalance issues and data scarcity prevalent across different regional dialects.

In specialized downstream NLP tasks, the complexity of dialectal morphology continues to pose significant hurdles. Kundu et al. [9] introduced *ANUBHUTI*, a dedicated corpus targeting sentiment analysis in regional languages, revealing that emotion markers vary significantly across regional boundaries. For the crucial task of offensive speech identification, Fayaz et al. [6] developed *BIDWESH*, evaluating hate speech detection mechanisms across regional dialects and proving that SCB-trained toxicity classifiers fail to generalize to dialectal slang. Concurrently, Paul et al. [14] formulated *ANCHOLIK-NER*, providing essential named entity annotations across regional variants to facilitate information extraction in non-standard texts.

2.2 Dialectal Machine Translation and LLM Benchmarks

While text classification laid the groundwork, direct efforts in generative machine translation for Bangla dialects have emerged only recently, driven by the advent of massively multilingual neural models. Early translation corpora were highly localized. For instance, Chowdhury et al. [4] built *ChatgaiyyaAlap*, a dedicated parallel corpus strictly designed for translating Chittagonian dialect sentences to Standard Bangla. To expand the dialectal scope, Sultana et al. [19] presented *ONUBAD*, providing an initial generalized resource for automated regional dialect conversion. This was followed by Faria et al. [5], who introduced *Vashantor*, a multilingual benchmark covering five regional dialects. Though a significant step forward, *Vashantor* remained constrained by its limited volume, offering only 2,500 sentences per dialect.

More recently, the research community has pivoted towards exploring the zero-shot and few-shot capabilities of Large Language Models (LLMs) for dialect translation. Mahjabin et al. [12] constructed *BanglaCHQ-Prantik* to benchmark state-of-the-art LLMs against human-evaluated dialect translation standards. Similarly, Jawad et al. [8] presented *DIALTSA-BN*, evaluating LLM capabilities in joint dialect translation and sentiment analysis. Furthermore, Bhuiyan et al. [3] explored *BhasaBodh*, investigating methods to bridge Bangla dialects and romanized forms through advanced machine translation techniques. Despite these advancements, fine-tuned NMT models consistently outperform generalized LLMs in specialized low-resource dialectal translation due to the LLMs' inherent lack of dialect-specific pre-training data.

2.3 Critical Synthesis and Research Gap

While the aforementioned contributions have collectively laid a valuable foundation for Bangla dialectal NLP, a critical review reveals several major structural and methodological shortcomings that currently stifle the development of robust, production-ready translation systems:

1. **Single-Dialect Isolation:** Corpora such as *ChatgaiyyaAlap* focus exclusively on a single dialect pair (e.g., Chittagonian-Standard). Models trained on such isolated datasets fail entirely to generalize across other regional dialects, necessitating the training of separate, disjoint models for every region.

2. **Severe Corpus Volume Limitations & Lack of Scaling Studies:** Multi-dialect datasets like *Vashantor* offer only modest data volumes (e.g., 2,500 samples per dialect). This scale is vastly insufficient for deep fine-tuning of architectures like mT5 or mBART-50. Furthermore, prior literature entirely lacks systematic scaling studies to empirically determine optimal dataset size thresholds for cross-dialectal transfer.

3. **The Pivot Translation Problem:** Existing works lack a unified framework capable of translating directly between different regional dialects (e.g., Sylheti $\rightarrow$ Chittagonian). Current systems force a "pivot" approach, routing translation through Standard Bangla as an intermediary, which compounds translation errors and degrades final semantic accuracy.

4. **Absence of Real-World Deployment:** State-of-the-art models developed in prior studies remain confined to offline evaluation environments. The community lacks open, deployed web applications that can provide real-time translation services directly to marginalized dialect speakers.

To definitively resolve these limitations, our work introduces a comprehensive, balanced, poly-dialectal parallel corpus that is orders of magnitude larger than previous efforts. By fine-tuning state-of-the-art Transformer models on this expansive dataset, we enable direct, multi-directional, poly-dialectal translation within a single, unified neural architecture.

2.4 Summary of Dialectal Resources

Table 1 summarizes the major dialectal datasets and resources reviewed in this section, detailing their coverage, architecture, evaluation metrics, and state-of-the-art performance.

Table 1: Comprehensive Summary of Bangla Regional Dialect NLP Resources and Datasets

Dataset & Year	**Primary Task & Domain**	**Dialects & Dataset Scale**	**Methodology & Quality Control**	**Key Metrics & SOTA Findings**
1. ANCHOLIK-NER (2025)	Dialectal Named Entity Recognition (NER); 10 entity classes.	Sylhet, Chittagong, Barishal. 10,443 sentences (3,481/dialect).	BIO tagging. Sourced from Vashantor (71.8%), ONUBAD, & news. 3-annotator consensus pipeline.	First dialectal NER baseline. Assamese NER baseline reached 80.69% F1 (MuRIL); medical BanglishBERT got 56.13% F1.
2. ANUBHUTI (2026)	Thematic Sentiment & Multilabel Emotion Analysis in dialectal text.	Mymensingh, Noakhali, Sylhet, Chittagong. 10,000 sentences (2,500/region).	Sourced from MONOVAB. 8 native translators (2/region) with 4-stage QA (sentiment, NaN parsing, anomaly, spelling).	High inter-rater agreement (Cohen's $\kappa = 0.76$–0.84). Dominant emotions: Contempt (921), Disgust (871), Anger (732).
3. BIDWESH (2025)	Multi-Dialectal Hate Speech & Toxicity Detection.	Barishal, Noakhali, Chittagong. 9,183 instances (3,061 parallel/dialect).	Sourced from BD-SHS; translated by 6 native speakers; 5-stage validation. Target/Type taxonomy.	Balanced binary split (49.2% Hate / 50.8% Non-Hate). Individual targets highest (34.4%); Slander dominant (53.7%).
4. Bangla Dialect Bibliography	Systematic Bibliography & Meta-Review of Bangla Dialect NLP.	Comprehensive overview across major regional variants in BD.	Systematic synthesis tracking 11 publications, annotation strategies, & baselines.	Mapped 11 unique corpora; highlights Sylheti & Chittagonian as most studied; Barishal, Mymensingh, Noakhali emerging.
5. BanglaDial (2025)	Dialect Identification (DID) & Class Imbalance Modeling.	11 Dialects + Standard Bangla. 60,729 sentences (362,430 words).	Aggregated from 4 corpora (Vashantor, ONUBAD, etc.). Deduplicated (4.1% removed) & normalized.	Long-tail distribution (Top-3: 41.3% of records). Imbalance Ratio $IR = 9.82$; Shannon Entropy $H = 3.416$ bits.

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Dataset & Year	Primary Task & Domain	Dialects & Dataset Scale	Methodology & Quality Control	Key Metrics & SOTA Findings
6. BdRegionText (2022)	Bangla Regional Text Classification on social media content.	Chittagong, Noakhali, Barishal, Rangpur. 2,573 sample texts.	Scraped from FB/YouTube/IG with human validation. Extracted TF-IDF, CountVectorizer, & BoW.	Random Forest + TF-IDF achieved peak 79.15% accuracy (10-fold CV: 79.47%, weighted F1: 0.78). RF + BoW got 75.92%.
7. BhasaBodh (2025)	Script Normalization & Multilingual Dialect Translation (Native/Romanized).	Chittagong, Sylhet (aligned with Standard Bangla & English). 1,960 sentences.	Sourced from ONUBAD; augmented via Gemini 2.5 Pro for Romanizations. Fine-tuned NLLB-200 & mBART-50.	mBART-50 achieved peak 87.44 BLEU on Romanized → Standard Bangla and 74.36 BLEU on Chittagong ↔ Sylhet.
8. ChatgaiyyaAlap (2025)	Dialect Conversion (Chittagonian → Standard Bangla).	Chittagonian (aligned with Standard Bangla). 4,012 sentences + 1,500-word dictionary.	Sourced from social media/dramas; validated by 5 native Chittagonian graduates.	Created clean translation mapping resolving severe grammatical divergences & phonetic shifts in negative sentences.
9. BanglaCHQ-Prantik (2025)	Medical-Domain Machine Translation (Standard Bangla → Dialects).	Sylheti (2,350 sentences), Chittagonian (500 sentences).	Sourced from BanglaCHQ-Summ; validated by 11 validators & 2 physicians. Evaluated LLMs (Qwen, Gemma, GPT-4o, Gemini).	Gemini 2.5 Flash achieved SOTA on Sylheti (23.67 BLEU); GPT-4o with CoT achieved SOTA on Chittagonian (24.67 BLEU).
10. DIALTSA-BN (2025)	Dialect Translation & Sentiment Analysis (Native vs. Transliterated).	Chattogram, Barishal, Sylhet, Noakhali. 600 annotated instances (150/region).	Sourced from YouTube; filtered via Word2Vec. Evaluated Claude 3.7, GPT-4o-mini, Gemma-3, Llama-3.1, etc.	Transliteration boosted translation BLEU from ¡0.07 to 0.330 (Claude 3.7) & 0.317 (GPT-4o-mini). Few-shot sentiment F1 reached 0.98.

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Dataset & Year	**Primary Task & Domain**	**Dialects & Dataset Scale**	**Methodology & Quality Control**	**Key Metrics & SOTA Findings**
11. BanglaCHQ-Prantik Baseline (2025)	Pediatric & Adult Clinical MT Error Taxonomy.	Sylheti, Chittagonian. 2,850 total aligned queries.	Parallel clinical reference tracking. Highlighted phonetic shifts (e.g., *pete* → *feto* / *cano*).	Identified 4 main LLM failure modes: terminology mismatch, dosage corruption (”500mg” → ”5 gram”), idioms, & orthographic drift.
12. ONUBAD (2025)	Fine-Grained Dialect Conversion & Neural Machine Translation.	Chittagong, Sylhet, Barisal. 7,950 total parallel rows (980 sentences/region).	Sourced from literature/volunteers; validated by university linguists. Benchmarked Seq2Seq & Transformers.	Documented 2 words/clause & 3 words/sentence. BanglaT5 and SeamlessM4T established peak NMT baselines.
13. Vashantor (2023)	Multi-Task Dialectal NMT & Region Classification.	Chittagong, Noakhali, Sylhet, Barishal, Mymensingh. 32,500 total sentences.	Sourced from social media/news; validated with Cohen’s & Fleiss’ Kappa. Custom DialectBanglaT5 & Dialect-BanglaBERT.	DialectBanglaT5 achieved SOTA on Mymensingh (71.93 BLEU). DialectBanglaBERT reached 89.02% region classification accuracy.

3 Research Hypothesis and Objectives

3.1 Research Hypothesis

We posit the following hypothesis: *A neural machine translation model trained on a unified, multi-dialectal parallel corpus encompassing multiple Bangla regional variants will demonstrate significantly improved generalization and cross-dialectal transfer compared to models trained on isolated, uni-dialectal datasets.* Specifically, we hypothesize that a poly-dialectal training paradigm exploits shared morphological and syntactic structures across related dialects, enabling the model to learn transferable representations that benefit even low-resource dialect pairs.

3.2 Research Objectives

The specific objectives of this research are:

1. To construct the largest multi-dialect parallel corpus for Bangla machine translation by integrating, aligning, and deduplicating seven existing dialectal datasets across 12 regional variants.

2. To develop a unified poly-dialectal translation system capable of three translation paradigms: Dialect → Standard Bangla, Standard Bangla → Dialect, and direct Dialect → Dialect translation.

3. To systematically evaluate and compare the performance of three state-of-the-art multilingual NMT architectures—BanglaT5 [2], NLLB-200 [22], and mBART-50 [21]—under parameter-efficient LoRA [7] fine-tuning on the constructed corpus.

4. To quantitatively analyze cross-dialectal transfer effects by examining how training data volume and linguistic proximity jointly influence translation quality.

5. To promote digital inclusion and linguistic accessibility for Bangla dialect speakers by establishing open benchmarks for future research.

4 Methodology

4.1 Dataset Construction and Integration

The foundation of our poly-dialectal NMT system is a meticulously curated multi-dialect parallel corpus. We integrate seven publicly available dialectal datasets, each contributing varying levels of coverage across different regional variants:

Table 2: Source corpus contributions per dialect. Values indicate the number of parallel sentence pairs available from each source.

Source Corpus	SCB	Syl	Cht	Bar	Mym	Noa	Ran	Raj
Ancholik-NER	3,481	3,481	3,481	3,481	3,481	3,481	0	0
Anubhuti	2,500	2,500	2,500	0	2,500	0	0	0
BanglaDial	3,452	442	577	790	712	0	655	891
BhasaBodh	980	980	980	0	0	0	0	0
ChatgaiyyaAlap	4,011	0	4,011	0	0	0	0	0
ONUBAD	980	980	980	980	0	0	0	0
Vashantor	2,500	2,500	2,500	0	2,500	2,500	0	0
Total	**13,556**	**6,422**	**10,567**	**4,270**	**5,712**	**5,000**	**1,155**	**891**

The final integrated dataset comprises 14,552 rows across 12 dialect columns (Standard Bangla plus 11 regional variants: Sylheti, Chittagonian, Barisali, Mymensingh, Noakhali, Rangpuri, Rajshahi, Kishoreganj, Narail, Narsingdi, and Tangail), yielding 51,531 total non-null parallel sentence pairs. This represents the largest such corpus for Bangla to date. As depicted in Figure 2, the dataset exhibits significant class imbalance—Chittagonian has 10,567 entries while the four newly added dialects (Kishoreganj, Narail, Narsingdi, Tangail) each have approximately 500—reflecting the inherent disparity in available dialectal resources.

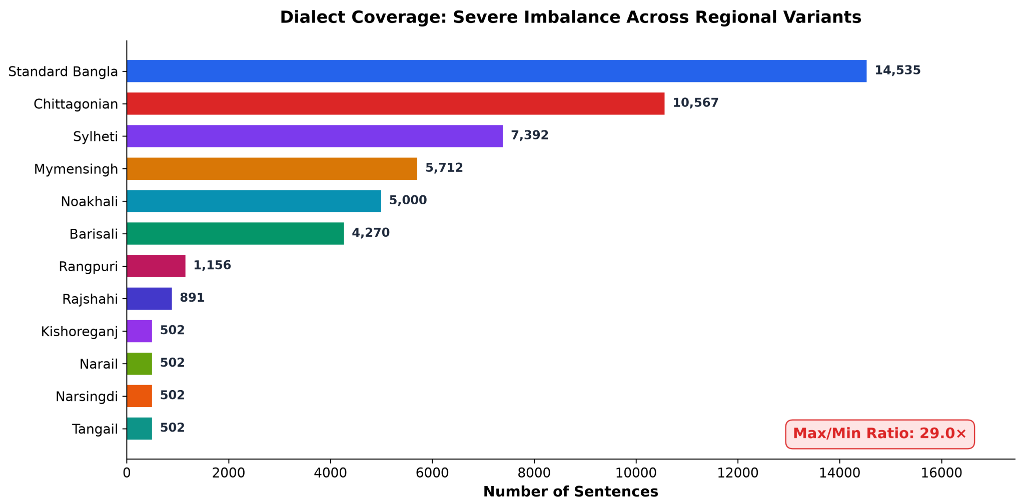

Figure 2: Dialect coverage distribution across the integrated corpus, revealing severe class imbalance between well-resourced dialects (Chittagonian, Sylheti) and underrepresented variants (Narail, Tangail).

4.2 Data Preprocessing Pipeline

The raw integrated corpus undergoes a rigorous four-stage preprocessing pipeline:

1. **Unicode Normalization:** All text is normalized to Unicode NFC form to resolve inconsistent character compositions prevalent in Bangla script.

2. **Cleaning and Deduplication:** Duplicate entries across source datasets are identified and removed. Empty, whitespace-only, and non-Bangla script entries are filtered.

3. **Tokenization and Script Alignment:** Text is tokenized using model-specific SentencePiece tokenizers. For BanglaT5, the 32K vocabulary is optimized for Bangla NLG tasks; for NLLB-200 and mBART-50, the larger vocabularies (∼256K and ∼250K, respectively) accommodate multilingual coverage.

4. **Train/Dev/Test Split:** The corpus is partitioned into training (80%), development (10%), and test (10%) sets with stratified sampling to ensure proportional dialect representation across splits.

4.3 Model Architectures

We evaluate three state-of-the-art encoder-decoder Transformer [23] architectures, chosen to represent a spectrum of design philosophies for multilingual and Bangla-specific NMT:

Table 3: Architectural comparison of the three NMT models evaluated in this study.

Feature	BanglaT5	NLLB-200	mBART-50
Parameters	247M	615M	611M
Architecture	T5	Transformer	BART
Encoder/Decoder Layers	12/12	12/12	12/12
Hidden Dimension	768	1,024	1,024
Attention Heads	12	16	16
FFN Dimension	3,072	4,096	4,096
Vocabulary Size	32K	∼256K	∼250K
Languages Supported	Bangla	200	50

BanglaT5 [2] is a T5-based [16] model pre-trained on approximately 27.5 GB of Bangla text, making it the only language-specific model in our comparison. **NLLB-200** [22] (No Language Left Behind, distilled 600M variant) is Meta AI's massively multilingual model designed for 200 languages. **mBART-50** [21] is Facebook AI's multilingual denoising auto-encoder based on BART [10] covering 50 languages.

4.4 Fine-Tuning Strategy

All models are fine-tuned using **Low-Rank Adaptation (LoRA)** [7], a parameter-efficient method that injects trainable low-rank decomposition matrices into each attention layer while freezing the pre-trained weights. We investigate two LoRA configurations:

- **Configuration A:** rank $r = 8$, scaling factor $\alpha = 16$, trained for 10 epochs.

- **Configuration B:** rank $r = 64$, scaling factor $\alpha = 128$, trained for 20 epochs.

The best-performing model (BanglaT5) is additionally trained for 100 epochs under Configuration A to examine the effect of extended training on convergence and performance saturation.

4.5 Multi-Directional Translation Paradigm

Unlike prior uni-directional approaches, our system is trained to handle all possible translation directions within a single unified model:

- **Dialect → Standard Bangla:** Converting regional dialect text to SCB.
- **Standard Bangla → Dialect:** Generating dialectal text from SCB input.
- **Dialect → Dialect:** Direct translation between two different regional dialects without an intermediary pivot through Standard Bangla.

4.6 Evaluation Metrics

Model performance is assessed using four complementary automatic evaluation metrics:

- **BLEU** [13] (Bilingual Evaluation Understudy): Measures n-gram precision between generated and reference translations.
- **chrF++** [15]: Character-level F-score incorporating word-level bigrams, robust to morphological variations.
- **METEOR** [1]: Harmonic mean of precision and recall with synonym matching and stemming.
- **TER** [18] (Translation Edit Rate): Minimum number of edits required to transform the output into the reference (lower is better).

5 Experimental Results and Analysis

5.1 Baseline Model Comparison

Table 4 presents the overall performance of the three NMT models after fine-tuning on the full poly-dialectal corpus. BanglaT5 consistently outperforms the two larger multilingual models despite having only 247M parameters—less than half the size of NLLB-200 (615M) and mBART-50 (611M).

Table 4: Overall performance comparison of the three NMT models on the poly-dialectal test set (18,062 translation pairs across all dialect directions). Best results are shown in **bold**.

Model	Params	BLEU ↑	chrF++ ↑	METEOR ↑	TER ↓
BanglaT5 (10 ep)	247M	23.22	51.20	41.41	56.85
NLLB-200	615M	14.76	42.62	33.56	64.53
mBART-50	611M	8.96	32.94	23.82	77.10
BanglaT5 (100 ep)	247M	**29.26**	**57.26**	**49.68**	**50.59**

The extended 100-epoch BanglaT5 training yields a 26% relative improvement in BLEU (from 23.22 to 29.26) over the 10-epoch baseline, demonstrating that dialect translation benefits substantially from prolonged fine-tuning. Figure 3 visualizes this comparison across all four metrics.

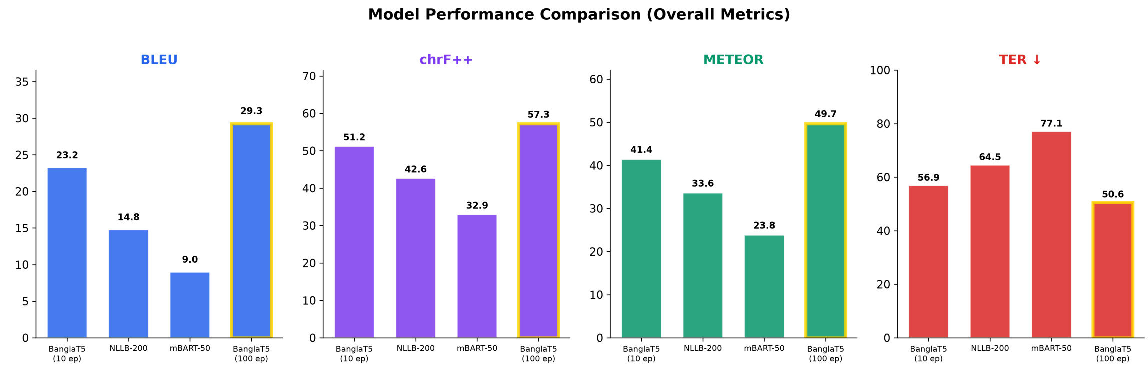

Figure 3: Comparative evaluation of four model configurations across BLEU, chrF++, METEOR, and TER. Gold-bordered bars indicate the best-performing configuration.

5.2 Cross-Dialectal Translation Analysis

Figure 4 presents a heatmap of BLEU scores for all source-target dialect pairs under the best model (BanglaT5, 100 epochs). Several key findings emerge:

1. **Morphologically proximate dialects translate better:** Mymensingh → Standard Bangla achieves the highest BLEU score of 55.0, consistent with Mymensingh's closer lexical similarity to SCB. Conversely, Rajshahi and Rangpur—dialects with distinct phonological systems—yield BLEU scores below 12 in most directions.

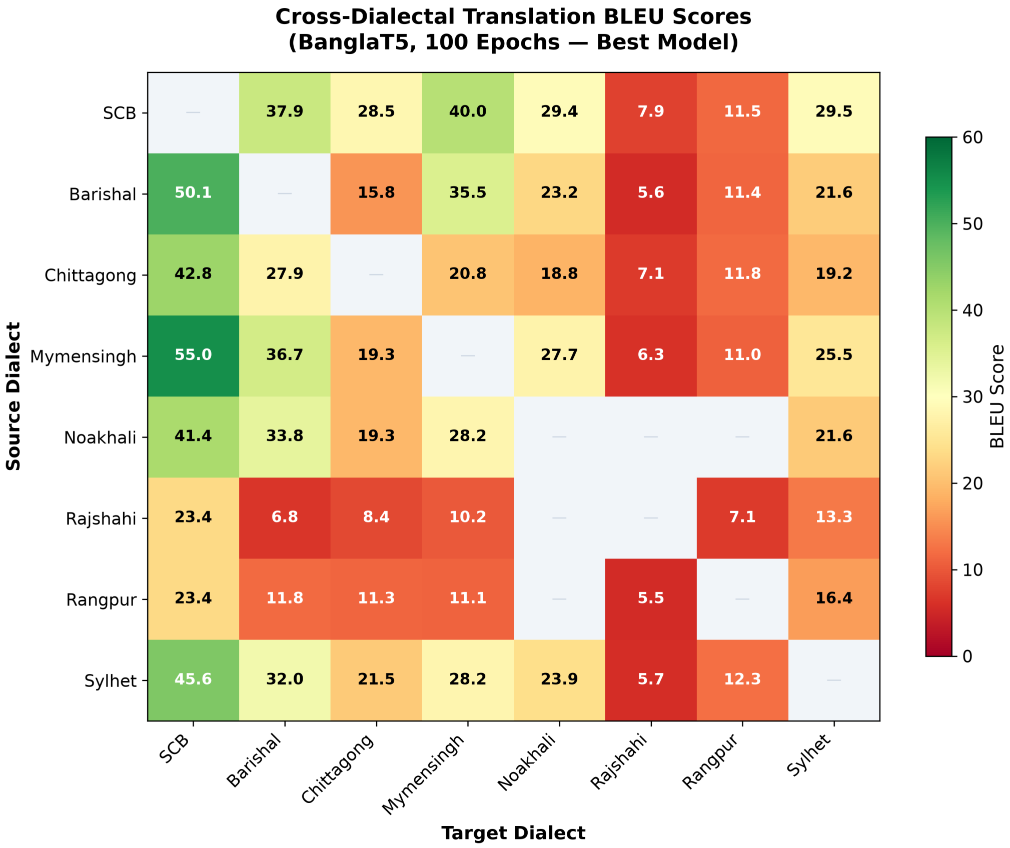

Figure 4: Cross-dialectal BLEU score heatmap (BanglaT5, 100 epochs). Darker green indicates higher translation quality. Rajshahi and Rangpur (bottom rows) exhibit uniformly low scores due to limited training data.

2. **Translation directionality is asymmetric:** As shown in Figure 5, translating *from* a dialect to Standard Bangla is consistently easier than the reverse. For example, Barisali → SCB achieves 50.1 BLEU while SCB → Barisali achieves only 37.9 BLEU, a gap of 12.2 points. This asymmetry reflects the decoder's greater difficulty in generating dialectal morphological forms absent from its pre-training distribution.

3. **Data volume directly correlates with performance:** Dialects with fewer than 1,000 training samples (Rajshahi: 891, Rangpur: 1,155) perform drastically worse than those with 5,000+ samples, underscoring the critical role of corpus volume in NMT quality.

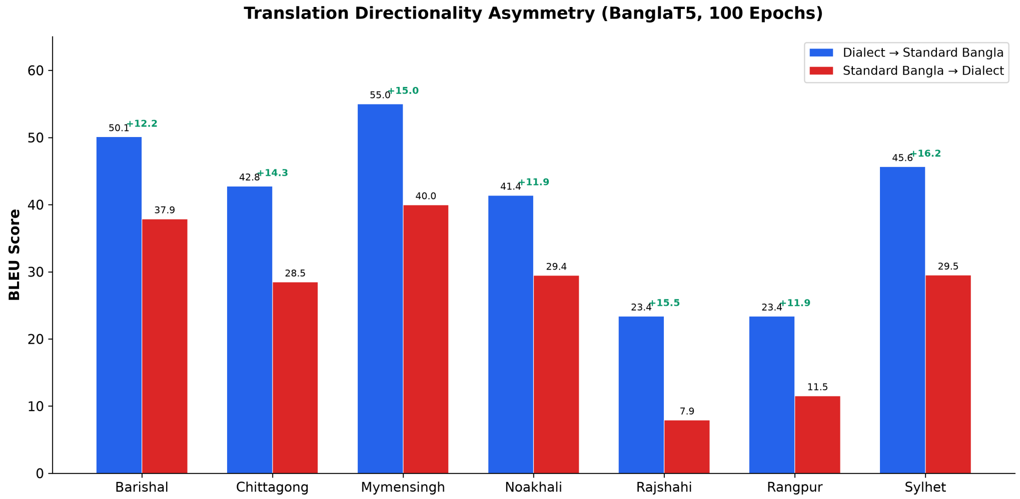

Figure 5: Translation directionality asymmetry: Dialect→SCB consistently outperforms SCB→Dialect, with green labels indicating the BLEU gap.

5.3 Dataset Scaling Analysis

To quantify the relationship between training data volume and translation quality, we conduct a controlled scaling experiment using BanglaT5 with LoRA fine-tuning. Training data is incrementally scaled from 500 to 4,499 sentence pairs per dialect combination, with results measured on a fixed test set of 10,000 pairs.

As shown in Figure 6, BLEU improves from 12.09 to 25.74 (a 113% relative increase) as data scales from 500 to 4,499 under the r=8 configuration. The higher-capacity r=64 configuration yields slightly better results (27.55 BLEU at 4,499 samples) but shows diminishing marginal returns beyond approximately 3,000 samples. This finding validates the central premise of this thesis: *larger, more diverse corpora are essential for effective poly-dialectal NMT.*

Figure 6: BLEU and chrF++ scaling curves as training data increases from 500 to 4,499 per dialect pair. Both LoRA configurations show consistent improvement with larger data, but with diminishing returns beyond 3,000 samples.

6 Limitations and Future Work

6.1 Current Limitations

1. **Persistent Class Imbalance:** Despite integrating seven source corpora, the dataset exhibits a 21:1 imbalance ratio between the most represented dialect (Standard Bangla: 14,535 entries) and the least represented (Kishoreganj, Narail, Narsingdi, Tangail: ∼502 each). This imbalance limits the model's generalization capacity for underrepresented dialects, as evidenced by the low BLEU scores for Rajshahi (23.4) and Rangpur (23.4) even under the best model.

2. **Computational Cost:** Fine-tuning even moderately sized models (247M–615M parameters) on the full multi-directional translation matrix requires substantial GPU memory and training time. The 100-epoch BanglaT5 training required approximately 851 seconds, while mBART-50 required 13,889 seconds—a 16× disparity highlighting the efficiency advantage of language-specific models.

3. **Text-Only Modality:** The current system operates exclusively on text input, neglecting the phonological richness and prosodic variation characteristic of spoken Bangla dialects. A speech-to-text pipeline would be necessary for real-world deployment in voice-based applications.

4. **Absence of Human Evaluation:** While automatic metrics (BLEU [13], chrF++ [15], METEOR [1], TER [18]) provide objective and reproducible quality assessments, they cannot fully capture translation fluency, dialectal authenticity, and semantic adequacy as perceived by native speakers.

6.2 Future Directions

1. **Corpus Expansion:** Systematic data collection for underrepresented dialects through community-driven annotation campaigns and web scraping of social media platforms.

2. **Multimodal Integration:** Extending the framework to incorporate speech-to-text and text-to-speech modules, enabling end-to-end dialectal speech translation.

3. **Advanced Training Strategies:** Investigating curriculum learning, data augmentation through back-translation, and multi-task training to improve performance on low-resource dialect pairs.

4. **Human Evaluation Protocol:** Designing and conducting a formal human evaluation study with native dialect speakers to assess translation adequacy, fluency, and dialectal authenticity.

7 Conclusion

This thesis has presented a comprehensive framework for poly-dialectal neural machine translation in Bangla, addressing the critical gap between the rich dialectal diversity of the language and the limited scope of existing NLP tools. The principal achievements of this work are threefold.

First, we constructed the **largest multi-dialectal parallel corpus** for Bangla machine translation to date, integrating seven existing datasets into a unified resource comprising 14,552 aligned entries across 12 regional dialect columns, totaling 51,531 parallel sentence pairs. This corpus introduces coverage for four previously unaddressed dialects (Kishoreganj, Narail, Narsingdi, and Tangail) and provides up to 2.76× more data per dialect than the best existing competitor.

Second, our systematic **model evaluation** revealed that BanglaT5—a language-specific model with only 247M parameters—decisively outperforms the larger multilingual models NLLB-200 (615M) and mBART-50 (611M), achieving a best overall BLEU of 29.26 and chrF++ of 57.26 after 100 epochs of LoRA fine-tuning. This finding underscores the importance of language-specific pre-training for low-resource intra-language translation tasks.

Third, our **cross-dialectal analysis** uncovered fundamental patterns in dialect translatability: (a) morphologically proximate dialects yield higher BLEU scores (Mymensingh→SCB: 55.0), (b) dialect-to-standard translation is consistently easier than the reverse by 10–15 BLEU points, and (c) translation quality scales log-linearly with training data volume, with diminishing returns observed beyond 3,000 samples per dialect pair.

By enabling direct, multi-directional translation without a pivot through Standard Bangla, this system represents a meaningful step toward inclusive, dialect-aware NLP infrastructure for the Bangla-speaking population. The corpus, models, and evaluation framework established in this thesis provide a robust foundation for future research in low-resource dialectal machine translation.

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